GROUP 7 - DOCUMENTATION
MILESTONE 1: DATA INGESTION & SETUP

1. Objective

The objective of this milestone is to set up the project environment, design the database schema, load raw CSV data into a SQLite database, and organize the workflow using Apache Airflow. This milestone forms the foundation for further data processing and transformation.

2. System Overview

The system performs data ingestion and storage using Python scripts and organizes the workflow using Apache Airflow.
Workflow:

- Raw CSV files are collected
- Python scripts process and load the data
- SQLite database is created
- Data is stored in structured tables
- Airflow DAG organizes execution flow (manual trigger)

This ensures that raw data is properly structured and ready for further ETL processing.

3. Tools and Technologies Used

Tool	Purpose
Python	Scripting and automation
Pandas	Data processing
SQLite	Database storage
Apache Airflow	Workflow orchestration

4. Project Setup

The project environment was initialized with a structured folder system.
Key Steps:

- Created project directory
- Organized scripts in scripts/ folder
- Installed required libraries (Pandas, SQLite, Airflow)
- Prepared raw CSV datasets

5. Database Creation

A SQLite database named **infosys_etl.db** was created.

Tables Created:

• orders
- order_id (Primary Key)
- order_date
- customer_name
- state
- city

• order_details
- order_id (Foreign Key)
- amount
- profit
- quantity
- category
- sub_category

• sales_target
- month
- category
- target

6. Scripts Used

6.1 create_tables.py

This script creates the database and tables.

Functionality:
- Connects to SQLite database
- Creates tables if not already present
- Defines schema with primary and foreign keys

6.2 load_data.py

This script loads raw CSV data into the database.

Functionality:
- Reads CSV files using Pandas
- Renames columns to match schema
- Removes duplicate records
- Inserts data into tables

Input Files:
- List of Orders.csv
- Order Details.csv
- Sales target.csv

6.3 check_data.py

This script validates the data loading process.
Functionality:

- Connects to database
- Counts number of rows in each table
- Displays output for verification

7. Data Ingestion Process

The ingestion process includes:

1. Reading raw CSV files
2. Standardizing column names
3. Removing duplicate entries
4. Loading data into SQLite tables

This ensures clean and consistent data storage.

8. Airflow DAG Implementation (Manual Execution)

An Apache Airflow DAG was created to organize the ETL workflow.
DAG Details:

- **DAG ID:** milestone1_etl_pipeline
- **Schedule:** None (manual execution)
- **Catchup:** Disabled

Tasks in DAG

The DAG consists of three tasks:

1. create_tables
   - Executes create_tables.py
   - Creates database tables
2. load_data
   - Executes load_data.py
   - Loads CSV data into database
3. check_data
   - Executes check_data.py
   - Validates data using row counts

Task Flow

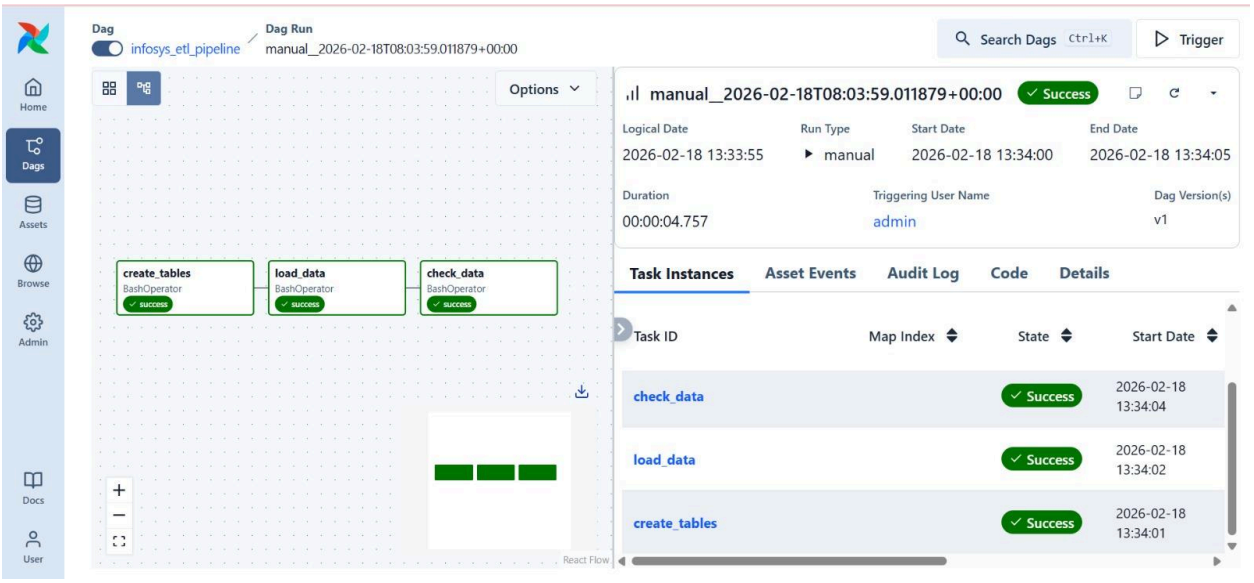
create_tables → load_data → check_data

Execution Method

The DAG is executed manually using the Airflow UI.
Steps:

1. Open Airflow Web Interface
2. Locate DAG: milestone1_etl_pipeline

3. Click **Trigger DAG**
4. Monitor execution status



Important Note
The DAG is configured for manual execution (schedule=None).
This means:

- No automatic scheduling
- Execution is user-triggered
- Useful for testing and controlled runs

9. Testing and Validation

The data ingestion process was validated using:

- Table creation verification
- Row count checking
- Data correctness validation

Validation Method:
Executed:
python check_data.py

Sample Output:
Total Orders: 500
Total Order Details: 1500
Total Sales Targets: 36

10. Execution Steps

To run the project:

1. Navigate to scripts folder:

```
cd milestone1/scripts
```

2. Run scripts in order:

```
python create_tables.py
python load_data.py
python check_data.py
```

11. Project Structure

```
milestone1/
│── scripts/
   ├── create_tables.py
   ├── load_data.py
   └── check_data.py
│── README.md
```

12. Conclusion

Milestone 1 successfully establishes the foundation of the project.

- Database schema was created
- Raw data was loaded into structured tables
- Data validation was performed
- Airflow DAG was implemented for workflow organization

This milestone prepares the system for data cleaning, transformation, and further analysis in upcoming stages.